

INPUT

Travel demand

Inter-zonal origin-destination
freight flows (tonnes)

Vehicle types | drivetrain technologies | fuels

Graph topology

Nodes (NUTS-2 regions) | edges
(corridor connections) | route definitions

Parameters on fueling infrastructure expansion

Differentiated by fuel & fueling power
| Maximum possible speed of build-up

Fuel prices | technology costs | fu-
eling infrastructure costs | route-
related / location-specific costs...

Battery & charging parameters

Battery capacity | charging power
| initial & final SOC constraints

Mandatory break regulations

EU driving time rules | minimum cu-
mulative travel times at break locations

Modal shift parameters

Maximum rail share | rate-of-change
constraints | rail infrastructure limits

Geographically varying costs

Market-zone electricity prices |
location-specific network fees

Optimization model

Core formulation:

- Linear program
- System cost-minimization
- Network-based transport demand coverage
- Vehicle stock modeling
- Sizing of fueling infrastructure
- Spatially flexible fueling activity

RQ3 extensions:

- State of charge tracking
- Travel time & mandatory break accounting
- Modal shift to rail
- Geographically varying charging costs

OUTPUT

Freight trips per drivetrain
technology & fuel

Vehicle stock shift

Fueling demand
by location

Fueling infrastructure
expansion

Charging load distribution
by market zone

State of charge profiles
along routes

Modal split
(road vs. rail)

Travel time profiles
along routes